

Master Dynamic Calculations with WINDOW, OFFSET & INDEX in DAX! 💡

Power BI's latest DAX functions are transforming how we build time intelligence and row-level calculations. If you haven't explored **WINDOW**, **OFFSET**, and **INDEX** yet, you're missing out on some serious analytical power!

Why These Functions Matter

These three functions work together to create flexible, dynamic calculations over datasets—perfect for moving averages, period-over-period comparisons, and custom aggregations without complex filter contexts.

WINDOW: Define Your Range

The **WINDOW** function creates a dynamic subset of rows based on position and order.

Example: 3-Month Moving Average

```
3-Month Moving Avg =  
AVERAGEX(  
    WINDOW(-2, 0, ORDERBY(Sales[Date], ASC)),  
    Sales[Amount]  
)
```

What it does: For each row, calculates the average of the current row and 2 preceding rows, ordered by Date.

OFFSET: Navigate Row by Row

The **OFFSET** function retrieves values from rows at a specific distance from the current row.

Example: Month-over-Month Growth

```
MoM Growth % =  
VAR CurrentSales = Sales[Amount]  
VAR PreviousSales =  
    SELECTCOLUMNS(  
        OFFSET(-1, ORDERBY(Sales[Date], ASC)),  
        "PrevAmount", Sales[Amount]  
    )  
VAR PrevValue = MINX(PreviousSales, [PrevAmount])  
RETURN  
    DIVIDE(CurrentSales - PrevValue, PrevValue, 0)
```

What it does: Compares current month sales to the previous month by fetching the value one row back.

INDEX: Pinpoint Specific Positions

The **INDEX** function retrieves values from an exact position within an ordered set.

Example: Get the Highest Sale Date

```
Top Sale Date =  
SELECTCOLUMNS(  
    INDEX(1, ORDERBY(Sales[Amount], DESC), ORDERBY(Sales[Date], ASC)),  
    "Date", Sales[Date]  
)
```

What it does: Returns the date of the first row (highest sale) when sorted by Amount in descending order.

Real-World Use Case: Rolling 7-Day Revenue with Comparison

```
7-Day Rolling Revenue =  
SUMX(  
    WINDOW(-6, 0, ORDERBY(Sales[Date], ASC)),  
    Sales[Amount]  
)  
  
vs Previous Week =  
VAR Current7Day = [7-Day Rolling Revenue]  
VAR Previous7Day =  
    CALCULATE(  
        [7-Day Rolling Revenue],  
        OFFSET(-7, ORDERBY(Sales[Date], ASC))  
    )  
RETURN  
    Current7Day - Previous7Day
```

Key Benefits

- ☑ **Cleaner DAX:** Replace nested CALCULATE/FILTER logic with intuitive window operations
- ☑ **Better Performance:** Optimized for large datasets with minimal context transitions
- ☑ **Enhanced Flexibility:** Build custom time intelligence without date tables
- ☑ **Advanced Analytics:** Moving averages, rankings, and comparisons made simple